

Employment Protection and Consumption: Evidence from Italy

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Motivation

People care about job safety (Baghai et al. 2023):

- ▶ Rothwell and Crabtree 2019: 91% of Americans view job security as a crucial aspect of their employment and their life
- ▶ Censis 2025: 50% of Italian employees in the private sector would prefer a job in the public sector due to higher employment protection

Labour market reforms in Southern Europe after 2008 (OECD 2016):

- ▶ reduction in firing costs for firms
- ▶ policymakers' aim: increase labour flexibility and firms' productivity
- ▶ workers' perspective: reduction in employment protection
⇒ employment risk ↑

This Paper (I)

“What is the impact of employment risk on household consumption?”

Two challenges:

1. Employment risk is endogenous (Fuchs-Schündeln and Schündeln 2005)
 - ▶ workers choose occupations according to their risk preferences
 - ▶ need an exogenous variation for causality
2. Employment risk affects permanent income as well
 - ▶ need to *control* for permanent income (Y_P) but...
 - ▶ Y_P is not observable
 - ⇒ need a structural model to tell apart the *sole* effect of risk

This Paper (II)

How I address (1):

- ▶ leverage on a major labour market reform in Italy in 2015 known as the *Jobs Act* (JA)
- ▶ reduced employment protection for workers hired after March 6, 2015, by firms with 15+ employees ("large" firms)
- ▶ compare consumption of workers affected by the JA with workers under the previous regulation - called the *Chart of Labour* (COL)

How I address challenge (2):

- ▶ standard incomplete market model with **employment risk heterogeneity**
- ▶ use empirical results to **estimate** the structural parameters via MSM
- ▶ run the empirical analysis on simulated data, **controlling for** Y_P
- ▶ **disentangle** how much of the effect is purely attributable to empl. risk

Preview of the Results

Empirical Results:

- ▶ JAs spend 3% less than COLs on at-home food (relative to income)
- ▶ 7.9% difference in imputed consumption (Blundell et al. 2008)

Model Results:

- ▶ **Estimation:** $\hat{\gamma} = 1.73$ and $\hat{\beta} = 0.97$ (with CRRA preferences)
- ▶ **Simulation:** difference in consumption shrinks to 3.4% controlling for Y_P
⇒ employment risk accounts for roughly 40% of the total effect
- ▶ **Optimal UI:** More insurance to JAs reduces consumption inequality and increases welfare

Outline

1. Employment Protection in Italy
2. Data, Empirical Specification and Results
3. The Model
4. Conclusions

Related Literature

The impact of (income) risk on consumption/wealth:

- ▶ **Euler Equation estimation:** Dynan 1993, Bertola et al. 2005, Christelis et al. 2020, Crump et al. 2022, Sciacchetano 2024
- ▶ **structural models:** Hubbard et al. 1994, Carroll 1997, Gourinchas and Parker 2002, Cagetti 2003, Kaplan and Violante 2014, Low et al. 2010, Graves 2025
- ▶ **direct survey questions:** Guiso et al. 1992, Jappelli and Pistaferri 2025
- ▶ **natural experiments:** Kantor and Fishback 1996, Gertler and Gruber 2002, Fuchs-Schündeln and Schündeln 2005, Clark et al. 2022
- ▶ **contribution:**
 - ▶ exploit time-based discontinuity in employment protection
 - ▶ decomposition of the total effect in risk and Y_P effect

The effect of the *Jobs Act* on:

- ▶ **firms hirings decisions:** Boeri and Garibaldi 2019, Sestito and Viviano 2018
- ▶ **fertility:** De Paola et al. 2021, De Paola et al. 2024
- ▶ **mobility and on-the-job search:** Bertoni et al. 2023
- ▶ **access to mortgages:** Mistrulli et al. 2023
- ▶ **contribution:** unintended effect of the JA on consumption inequality

Employment Protection in Italy

Chart of Labour (1970):

- ▶ applies to workers of firms with 15+ employees hired until March 6, 2015
- ▶ **reinstatement** in case of unjust dismissal (same wage and positions)
- ▶ strong protection against unemployment

Jobs Act (2015):

- ▶ applies to workers hired from March 7 onwards
- ▶ reinstatement only in case of **discrimination** by the employer
- ▶ **graded security**: monetary compensation increasing with tenure
- ▶ **reduced protection** in the current occupation
 - ⇒ **higher** risk of becoming unemployed
 - ⇒ **higher** job displacement costs (Jarosch 2023, Bertheau et al. 2023)

Participation, Labour and Unemployment Survey (PLUS)

- ▶ 2014 and 2018 representative cross-sections of the 18-75 population
- ▶ information on firm size and hiring day (before/after March 7)
- ▶ tenure (in 5-year bins) and risk aversion (only in 2018 [See Question](#))
- ▶ at-home food spending
- ▶ sample characteristics: private sector employees with open-ended contracts

Household Budget Survey (HBS)

- ▶ detailed info on household spending from 2014 to 2018
- ▶ Italian counterparts of the CEX in the United States
- ▶ impute log total spending from HBS in PLUS (Blundell et al. 2008)

Labour Force Survey (LFS)

- ▶ look at hiring patterns before and after the reform
- ▶ net aggregate employment flows

Empirical Specification

I estimate the following equation, with employees of **large firms** in 2018.

$$Y_i = \beta \text{Jobs Act}_i + \varphi \mathbf{X}_i + \gamma_{\tau(i)} + \theta_{s(i)} + \varepsilon_i \quad (1)$$

Where:

- ▶ $Y_i \in \{(\text{food}/\text{income})_i, \log_{BPP}(c_i)\}$
- ▶ γ_{τ} (bins of 5 years) and θ_s are tenure and sector FEs
- ▶ $\mathbf{X}_i$ are controls (age, education, gender, family size, civil status...)
- ▶ Jobs Act_i dummy equal to 1 if hired starting from March 7, 2015
- ▶ **Identification Assumption:** workers do not choose the hiring date
 $\Rightarrow \text{Jobs Act}$ as good as randomly assigned conditioning on X

Main Results: Food over Income

Dependent Variable: Food over Family Income $\times 100$

	(1)	(2)	(3)	(4)
<i>Panel A: North of Italy</i>				
Jobs Act	-1.68** (0.68)	-1.75** (0.68)	-2.71** (1.21)	-2.76** (1.23)
Observations	2262	2262	2262	469
<i>Panel B: South and Centre of Italy</i>				
Jobs Act	0.74 (1.03)	0.59 (1.05)	0.68 (1.57)	0.55 (1.63)
Observations	1461	1461	1461	304
Controls	✓	✓	✓	✓
Job Sector FE		✓	✓	✓
Tenure			✓	≤ 5 years

Notes: Standard errors, clustered at the individual level, in parentheses. * $p < 0.10$,

** $p < 0.05$, *** $p < 0.01$

Main Results: BPP Imputation

[Details on the Imputation](#)

Dependent Variable: BPP-Imputed Log of Spending $\times 100$

	(1)	(2)	(3)
Jobs Act	-8.58** (3.58)	-8.68** (3.59)	-7.91** (3.62)
Observations	3753	3753	3753
Job Sector Dummies		✓	✓
RA Dummies			✓

Notes: Standard errors, clustered at the individual level, in parentheses. * $p < 0.10$,

** $p < 0.05$, *** $p < 0.01$

- ▶ difference in imputed consumption between 8 and 9%
- ▶ column 3: controlling for risk aversion, results are practically unchanged

Placebo Tests

- ▶ No differences in income between JAs and COLs [See](#)
- ▶ No effect on small firm and public sector employees [See](#)
- ▶ No anticipation of the reform [See](#)

Model Setup

- ▶ Time is infinite and discrete, a period is 1 year and no aggregate risk
- ▶ Two types of consumers/workers
 1. safe employees (mimic COL workers): low separation rate
 2. risky employees (mimic JA workers): high separation rate
- ▶ Employment risk regulated by Π_j with $j \in \{COL, JA\}$
 - ▶ job finding rate f
 - ▶ job separation rate s_j , with $s_{JA} > s_{COL}$
- ▶ Productivity and Income
 - ▶ log of productivity modeled as a AR(1) with ρ and σ_ε
 - ▶ income when employed: wage rate $w \times$ productivity
 - ▶ income when unemployed: means-tested UI $b(p_t)$
- ▶ Households can save in one asset with a constant return r

Formal Model

$$\max_{\{a_{j,t+1}, c_{j,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{j,t}^{1-\gamma} - 1}{1-\gamma} \right] \quad j \in \{COL, JA\}$$

$$a_{j,t+1} = (1+r)a_{j,t} + y(e_t(e_{t-1}, \Pi_j), p_t) - c_{j,t}$$

$$a_{j,t+1} \geq 0$$

$$y(e_t(e_{t-1}, \Pi_j), p_t) = \begin{cases} w \times p_t & \text{if } e_t = 1 \\ b(p_t) & \text{if } e_t = 0 \end{cases}$$

$$b(p_t) = \begin{cases} b_0 \times p_t & \text{if } p_t \leq b_1 \times \bar{p} \\ b_1 \times \bar{p} & \text{if } p_t > b_1 \times \bar{p} \end{cases}$$

$$\log(p_t) = \rho \log(p_{t-1}) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

$$\Pi_j = \begin{bmatrix} 1-f & f \\ s_j & 1-s_j \end{bmatrix}$$

$$s_{JA} > s_{COL}$$

Estimation Results

[See Estimation Details](#)[See External Calibration](#)

Parameter	Description	Value	Std. Error
γ	Relative Risk Aversion	1.727	0.171
β	Discount Factor	0.967	0.0024
s_{COL}	Separation Rate for COLs	0.047	0.0009
s_{JA}	Separation Rate for JAs	0.123	0.00549
f	Job Finding Rate	0.413	0.016

Moment	Source	Model	Data
Consumption Difference	Empirical Estimate	-7.85	-7.91
Mean Wealth-to-Income Ratio	SHIW 2016	6.51	6.43
Median Wealth-to-Income Ratio	""	4.53	4.44
P90 Wealth-to-Income Ratio	""	14.69	14.58
Mean Years of Tenure of COLs	LFS 2018	19.17	18.30
Unemployment Rate	EUROSTAT 2016-2019	12.30	11.06
Long Run Unemployment	""	7.20	6.50

OLS with Simulated Data

[See Distributions](#)[See Simulated Incomes](#)

Dependent Variable: Log of Simulated Consumption (× 100)		
	(1)	(2)
Jobs Act	-7.849*** (0.116)	-3.356*** (0.107)
PDV of Income		1.575*** (0.004)
Initial Wealth		0.799*** (0.011)

- ▶ simulate 250 periods and drop the first 50 $\left[PDV_i \equiv \sum_{t=0}^{200} \frac{y_{it}}{(1+r)^t} \right]$
- ▶ column 1: drop in consumption of 7.85% (targeted!!)
- ▶ column 2: controlling for Y_P the difference is 3.4% $\simeq$ 40% of the total effect

UI that maximizes welfare

- ▶ What are the optimal (potentially type-specific) parameters of UI?
- ▶ following Pellegrini 2025 and Zanella Cavallero et al. 2026, I solve:

$$\max_{\Omega} \omega_{COL} \int_{COL} V_{COL}(S; \Omega) g_{COL}(S; \Omega) dS + \omega_{JA} \int_{JA} V_{JA}(S; \Omega) g_{JA}(S; \Omega) dS$$

s.t.

$$\omega_{COL} \left[(1 - empl_{t,COL}) \left(\int_{p_t} \min \{ b_{0,COL} p_t; b_{1,COL} p_{50} \} d\mu \right) \right] +$$
$$\omega_{JA} \left[(1 - empl_{t,JA}) \left(b_{0,JA} \int_{p_t} \min \{ b_{0,JA} p_t; b_{1,JA} p_{50} \} d\mu \right) \right] \leq \overline{T}(\Theta, \psi)$$

$$\Omega \equiv \{ b_{0,COL}, b_{1,COL}, b_{0,JA}, b_{1,JA} \}$$

- ▶ g is the distribution over states S and μ the productivity distribution
- ▶ maximize total welfare subject to the same fiscal cost as the baseline $\overline{T}$

Optimal UI: Results

Parameter	Description	Value
<i>Optimal (type-variant) UI Structure</i>		
$b_{0,COL}$	Proportional UI for COLs	0.67
$b_{1,JA}$	Cap of UI for COLs	0.51
$b_{0,COL}$	Proportional UI for COLs	0.83
$b_{1,JA}$	Cap of UI for JAs	0.63
<i>Baseline (type-invariant) UI Structure</i>		
b_0	Proportional Component	0.75
b_1	Cap Component	0.54

- ▶ more insurance to JAs and less to COLs is welfare-improving!
- ▶ consumption difference goes down by 2.5 percentage points to 5.5%
- ▶ UI benefits should reflect more actual employment risk
- ▶ **one caveat:** no moral hazard effects (employment decisions are exogenous)

Conclusion

- ▶ JA workers spend 3% less on food than COL workers
- ▶ 8% difference in imputed consumption
- ▶ SIM economy with ex-ante risk heterogeneity to rationalize the result
- ▶ sensible estimates of RRA and of the DF
- ▶ 40% of the effect attributable to risk
- ▶ providing more UI to JA workers (and less to COL) maximizes welfare

Thank You!!



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Appendix

Summary Statistics by contract type: only big firm employees

	JA		COL	
	Mean	Sd	Mean	Sd
Between 18 and 24	0.12	0.33	0.02	0.14
Between 25 and 29	0.20	0.40	0.05	0.23
Between 30 and 39	0.31	0.46	0.26	0.44
Between 40 and 49	0.24	0.43	0.37	0.48
Between 50 and 64	0.13	0.34	0.29	0.45
Male	0.63	0.48	0.66	0.48
Married	0.41	0.49	0.63	0.48
Family Size	2.99	1.22	3.02	1.18
South and Islands	0.20	0.40	0.17	0.38
Centre	0.20	0.40	0.21	0.41
North	0.60	0.49	0.62	0.48
Low Education	0.23	0.42	0.37	0.48
Middle Education	0.52	0.50	0.49	0.50
High Education	0.26	0.44	0.14	0.35
Food Share (%)	27.14	12.81	29.55	12.19
Food Consumption (€100)	6.49	4.12	6.74	4.00
Individual Net Income (€100)	13.43	5.33	14.01	5.22
Family Net Income (€100)	24.42	10.69	23.13	10.62
Weekly average hours worked	38.24	9.05	37.42	8.20

Question on Risk Aversion

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Imagine you are being offered a lottery ticket for which you have 50 % chance of winning your net annual income. Would you exchange the ticket for...

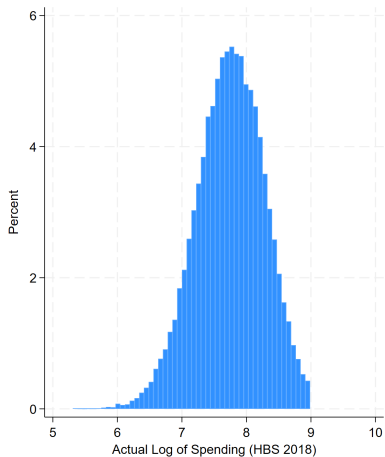
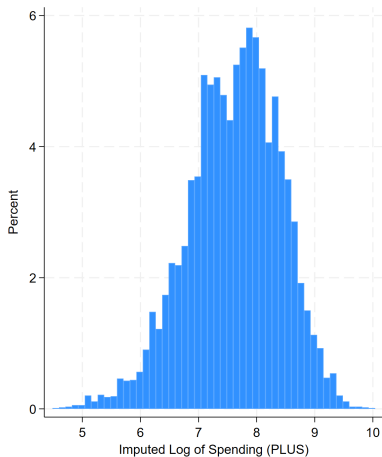
1. *5 % of your annual income*
2. *10 % of your annual income*
3. *25 % of your annual income*
4. *50 % of your annual income*
5. *75 % of your annual income*

- ▶ I estimate in HBS the following demand equation with OLS:

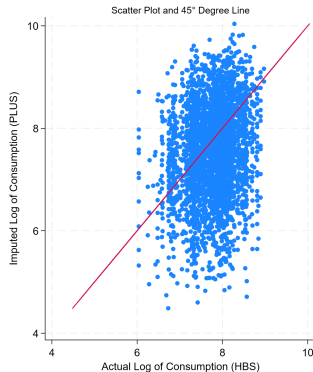
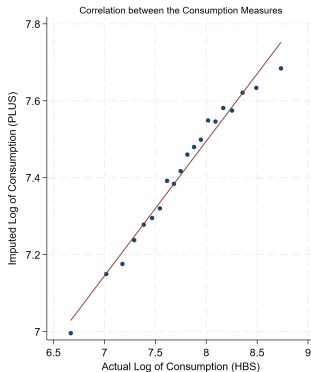
$$\log(f_i) = \alpha + \mu \mathbf{X}_i + \delta \log(c_i) + \varepsilon_i \quad (2)$$

- ▶ $\log(c_i)$ and $\log(f_i)$ are food and total spending
- ▶ δ is the elasticity of food spending w.r.t. total spending
- ▶ X_i are demographic characteristics common across PLUS and HBS
- ▶ Compute $\log_{BPP}(c_i)$ in PLUS using $\hat{\mu}$ and $\hat{\delta}$ [See Distribution](#) [See Binscatter](#)

BPP Imputation: Goodness of the Imputation (I) [Back](#)



BPP Imputation: Goodness of the Imputation (II)

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Difference-in-Differences [Back](#)

- ▶ Instead of eq (1), I could estimate the following D-in-D specification:

$$Y_{it} = \beta_{DID} Jobs\ Act_{it} \times Large_{it} + \delta Jobs\ Act_{it} + \gamma Large_{it} + \varphi \mathbf{X}_{it} + \lambda_t + \varepsilon_{it}$$

- ▶ Where:

- ▶ $Jobs\ Act_{it}$ is defined as above (hired after March 6, 2015)
- ▶ $Large_{it}$ dummy equal to 1 if working in a 15+ employees firm
- ▶ β_{DID} is the coefficient of interest

- ▶ Issues (Boeri and Garibaldi 2019 and Mistrulli et al. 2023):

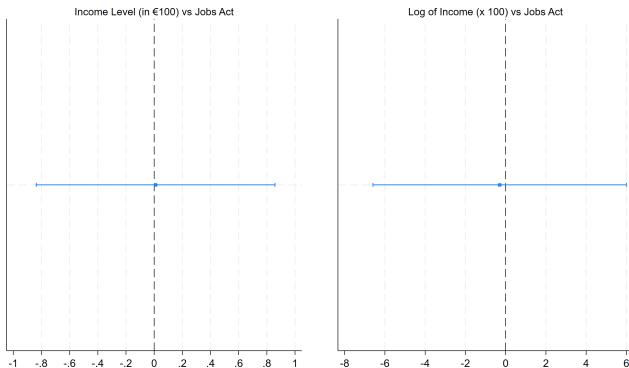
1. hiring subsidy for all firms starting from January 2015
 - ▶ distortion of wages
 - ▶ incentive for the marginal firm (14 employees) to hire one employee
 - ▶ firm dimension easily manipulable
2. selection into big/small firms driven by unobservables
 - ▶ selection in bigger firms $\Rightarrow$ negative bias (upper bound of the effect)
 - ▶ selection in smaller firms $\Rightarrow$ positive bias (lower bound of the effect)
3. not possible to check parallel trends (only 2014 before the reform)

Difference in Differences: Results [Back](#)

	Food Share (1)	Hours Worked (2)	Imputed (3)
Jobs Act x Large	-1.15 (0.83)	1.05 (0.71)	-12.03** (6.08)
Employed in a Big Firm	-1.46*** (0.26)	3.26*** (0.21)	12.53*** (1.85)
Jobs Act	-0.15 (0.74)	0.24 (0.62)	-0.03 (5.03)
Observations	12116	12608	11105

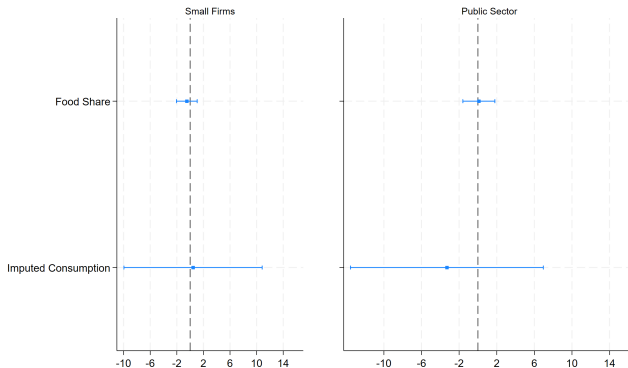
Notes: All regressions include a time dummy for 2018. Firms between 10 and 14 are excluded. Standard errors, clustered at the individual level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Placebo I: Regress Income vs Jobs Act

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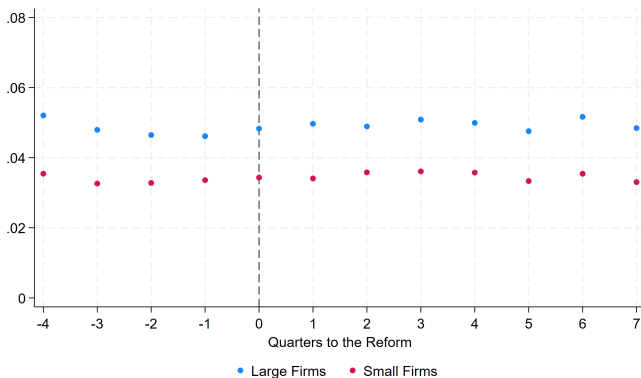
- ▶ no income differences between JA and COL workers
- ▶ variation in Y_P only coming from different separation rates

Placebo II: Small Firms and Public Sector

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- ▶ small firm and public sector worker not involved in the reform
- ▶ no economically and statistically significant effect

Placebo III: Hiring Rates Before around 2015-Q1

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- smooth hiring rates before and after the reform

Parameter	Description	Value	Source/Target
ρ	Persistence of AR(1)	0.95	Jappelli et al. (2024)
σ_{ε}	Std of the shock	0.50	Jappelli et al. (2024)
w	Wage Rate	1	Normalization
r	Annual Interest Rate	0.02	Jappelli et al. (2024)

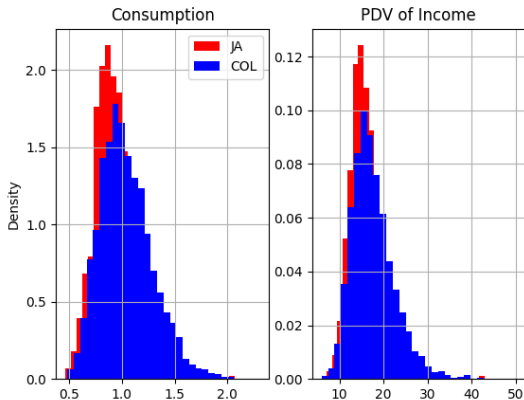
Parameters and Targeted Moments [Back](#)

I solve the following minimization problem:

$$\min_{\Theta} \left[m(\Theta; \psi) - \hat{m} \right]' W \left[m(\Theta; \psi) - \hat{m} \right]$$

- ▶ $\Theta = [\gamma, \beta, s_{COL}, s_{JA}, f]$
- ▶ ψ vector of exogenous parameters
- ▶ $m(\Theta; \psi)$ model-based moments, $\hat{m}$ data moments
- ▶ ψ is the set of exogenous parameters
- ▶ W is the inverse of the covariance matrix of moments
- ▶ bootstrap 100 samples of the micro data

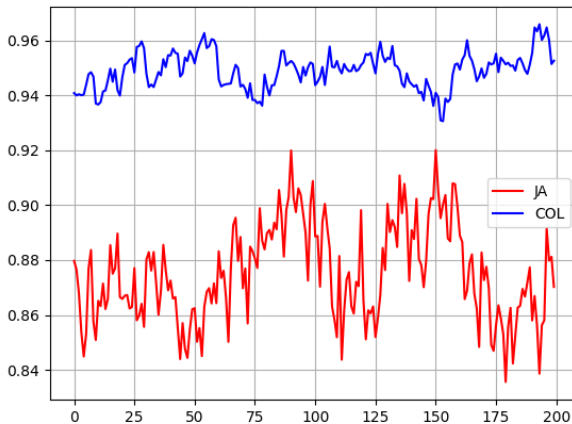
Consumption and PDV of Income

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► lower average consumption for JAs compared to COLs but...

► $\simeq 7\%$ lower PDV of income as well $\left[PDV_i \equiv \sum_{t=0}^{200} \frac{y_{it}}{(1+r)^t} \right]$

Simulated Income Paths

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- ▶ higher income for COL workers on average and...
- ▶ higher volatility for JA

Two Account Model: Setup

Calibration

- ▶ liquid asset b : r_b and adjustable every period
- ▶ illiquid asset a : $r_a > r_b$ but infrequent adjustment
- ▶ Auclert et al. 2024: χ iid. prob. of adjustment of a

$$\max_{\{a_{j,t+1}, b_{j,t+1}, c_{j,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{j,t}^{1-\gamma} - 1}{1-\gamma} \right] \quad j \in \{COL, JA\}$$

$$b_{j,t+1} + a_{j,t+1} + d_{jt} \mathbb{I}_{adj=1} = (1 + r_b) b_{j,t} + y_{jt} - c_{j,t}$$

$$a_{j,t+1} = (1 + r_a) a_{j,t} + d_{jt} \mathbb{I}_{adj=1}$$

$$b_{j,t+1} \geq 0, \quad a_{j,t+1} \geq 0$$

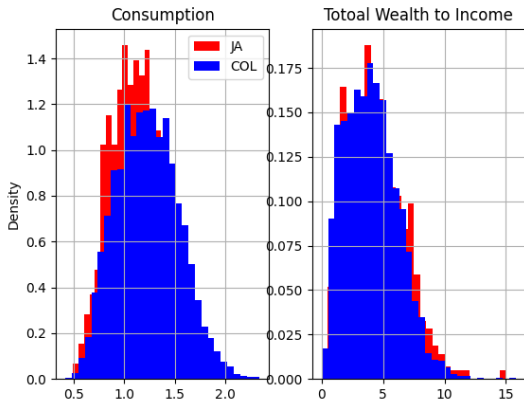
Two Account Model: Estimation

Parameter	Value	Moment	Model	Data
γ	2.23	Diff in Consumption	-7.89	-7.91
β	0.92	Illiquid Wealth to Income Ratio	6.42	6.44
χ	0.10	Average MPC	0.47	0.48
s_{COL}	0.05	COLs Years of Tenure	19.35	18.31
s_{JA}	0.15	Unemployment Rate	11.74	10.75
f	0.46	Long run Unemployment Rate	6.33	6.50

Two Account Model: Results

Differences between JA and COL

$\% \Delta a/y$	$\% \Delta b/y$	$\% \Delta (a+b)/y$	$\% \Delta c$
10.82	25.30	12.34	-7.32



EZ Preferences: Estimation Results

Now I assume preferences as in Epstein and Zin 1989

$$\max_{c_j, a_{jt+1}} V_{jt} = \left[(1 - \beta) c_{jt}^{\frac{\theta-1}{\theta}} + \beta \mathbb{E}_t \left[V_{jt+1}^{1-\gamma} \right]^{\frac{\theta-1}{\theta(1-\gamma)}} \right]^{\frac{\theta}{\theta-1}}$$

► θ is the *EIS* and γ is risk aversion

Parameter	Value	Std Error	Moment	Model	Data
γ	1.54	0.232	Log Diff. in Consumption	-8.23	-7.91
eis	0.29	0.0084	Mean Wealth to Income	6.48	6.43
β	0.95	0.0045	Median Wealth to Income	4.67	4.44
	-	-	P90 Wealth to Income	14.18	14.58
s_{COL}	0.05	0.0049	COLs Years of Tenure	17.85	18.30
s_{JA}	0.13	0.002	Unemployment Rate	11.12	11.06
f	0.47	0.012	Long Run Unemployment Rate	5.93	6.50

OLS with EZ Preferences

Dependent Variable:	Log of Simulated Consumption ($\times 100$)		
	(1)	(2)	(3)
Jobs Act	-7.886*** (0.116)	-3.415*** (0.108)	-6.364*** (0.388)
PDV of Income		1.565*** (0.004)	1.552*** (0.005)
Initial Wealth		0.703*** (0.011)	0.707*** (0.012)
Jobs Act $\times$ PDV of Income			0.102*** (0.013)
Jobs Act $\times$ Init. Wealth			-0.032 (0.030)